Paroxysmal nocturnal hemoglobinuria

Acquired PIGA-mutant clone · hemolysis, thrombosis, marrow failure · **diagnose by granulocyte FLAER flow, treat hemolysis or thrombosis with complement inhibition, vaccinate before starting**

Compiled from ICCS/ESCCA flow guidance · IPIG · FDA labels · primary trial reports verified against PubMed, October 2026.

PATHOPHYSIOLOGY

- **Somatic PIGA mutation in a stem cell → no GPI anchor → CD55 + CD59 absent** from every progeny cell → RBCs lysed by the alternative complement pathway. The clone expands only where the normal marrow is under immune attack — hence the overlap with aplastic anemia
- **Triad: intravascular hemolysis, thrombosis, marrow failure.** Free Hb scavenges NO → **dysphagia, abdominal pain, erectile dysfunction**, pulmonary hypertension, CKD. Dark morning urine is the name, not the rule
- **Untreated natural history:** median survival 10 yr; 39% had venous thrombosis *Hillmen NEJM 1995, n=80*. **On eculizumab, survival matches age-matched controls** *Kelly Blood 2011*

CLINICAL FEATURES

- **DAT-negative intravascular hemolysis** — high LDH, absent haptoglobin, hemoglobinuria or hemosiderinuria
- **Unexplained thrombosis at an unusual site** — Budd-Chiari, mesenteric/portal, cerebral venous sinus, dermal — or thrombosis + hemolysis or cytopenias
- **Unexplained cytopenias** — PNH is a marrow failure disorder: anemia with a **retic low for the degree of hemolysis**, or any cytopenia + hemolysis/thrombosis
- **Aplastic anemia: at diagnosis and periodically** (yearly, or with any sign of hemolysis); **hypoplastic MDS**
- **Do not use Ham or sucrose lysis** — obsolete. Do not send RBC CD55/CD59 alone

FLOW CYTOMETRY — ICCS/ESCCA

Lineage	Reagents · what it tells you
Granulocytes	**FLAER + CD24**, gated on CD15. **The clone-size number** — unaffected by hemolysis or transfusion
Monocytes	**FLAER + CD14**, gated on CD64. Confirms a second lineage
RBCs	**CD59**, gated on CD235a. **Type III** = absent (lyses), **type II** = partial, type I = normal. **Underestimates the clone** — PNH RBCs are destroyed and transfused cells dilute
Sensitivity	Routine ~1%. **High-sensitivity 0.01%** — needed for AA/MDS screening *Borowitz 2010; Illingworth 2018*

FLAER binds the GPI anchor itself, so it is lineage-agnostic — but it does not work on RBCs. Report % granulocyte, % monocyte and RBC type II + III separately.

MONITORING

- **Each visit:** CBC, retic, LDH, bili; DAT (C3) if anemic on a C5i; ferritin, B12, folate periodically
- **Flow clone size every 6–12 mo** (granulocytes) — and in AA/MDS patients without PNH, yearly screening
- Renal function; new cytopenias → marrow (AA or MDS evolution)

TREATMENT PATHWAY

Hemolytic PNH needing therapy

↓

1. Vaccinate before dose 1
MenACWY + MenB ≥2 wk before
proximal agents: + pneumococcal
urgent start: antibiotic cover until 2 wk post-vaccine

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2. Choose the agent
C5i: ravulizumab IV q8wk or crovalimab SC q4wk
— IVH + thrombosis control, longest record
Proximal: iptacopan PO or pegcetacoplan SC
— also fixes EVH (higher Hb); adherence-critical,
BTH can be severe. **Pregnancy → eculizumab/ravulizumab**

↓

3. Reassess at 3–6 months
Hb, LDH, retic, bili, transfusions, DAT (C3)

On a C5 inhibitor and still anemic

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Sort out why
LDH ≤1.5× ULN + high retic + DAT C3+ → **EVH**
LDH rising → **BTH** (trough or trigger) — page 2
low retic → **marrow failure**, iron, B12, folate

↓

Clinically significant EVH
switch to iptacopan (APPLY) or pegcetacoplan (PEGASUS), **or add** danicopan to the C5i (ALPHA)

Marrow failure dominates

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Treat the aplastic anemia
IST (hATG + cyclosporine ± eltrombopag)
or allo-HCT; complement Rx only for hemolysis

Allo-HCT

↓

Indications
severe aplastic anemia / MDS or AML evolution
refractory hemolysis or recurrent thrombosis
despite inhibition, or no access to inhibitors

WHO NEEDS TREATMENT

- **Treat:** symptomatic hemolysis (transfusion need, fatigue, smooth-muscle symptoms), **any thrombosis**, hemolysis-related organ damage (renal, pulmonary hypertension), **pregnancy**
- **Watch:** subclinical or small clones without hemolysis — CBC + LDH, repeat flow every 6–12 mo

BASELINE LABS

- **Hemolysis:** LDH (the IVH marker), retic, haptoglobin, indirect bili, free Hb, UA + **urine hemosiderin**. **DAT negative** (becomes C3d+ on a C5 inhibitor)
- **Iron, B12, folate:** chronic hemoglobinuria causes **iron deficiency**; on C5i, transfused EVH can swing to overload
- **CBC + marrow biopsy with cytogenetics** if any cytopenia — is there AA or MDS underneath? Creatinine; imaging only for symptoms

IPIG CLASSIFICATION — PARKER 2005

Category	Defines
Classic	Clinical IVH, cellular marrow, no other marrow disorder. Usually a large clone
PNH + another BMF	Hemolysis with AA or low-risk MDS documented — treat whichever dominates
Subclinical	Small clone, **no clinical or lab hemolysis**; seen in AA/MDS. **No complement therapy** — monitor the clone

THROMBOSIS

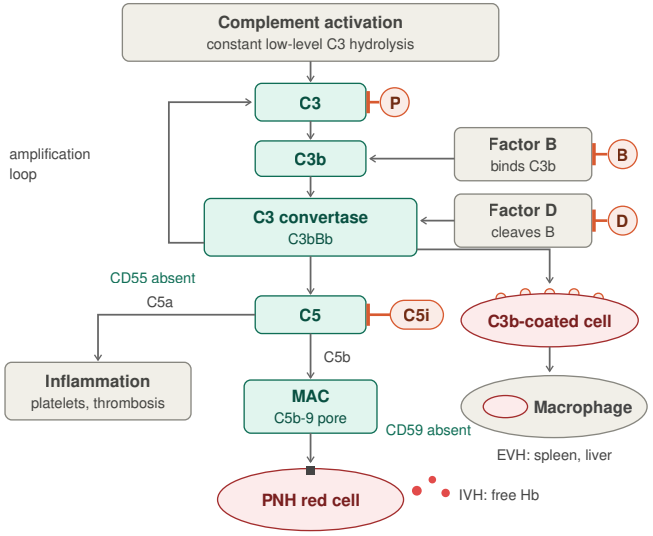
- **Leading cause of death** in PNH. **Mostly venous, often unusual sites**; can occur on therapeutic anticoagulation
- **Clone size predicts it:** granulocyte clone >50% → 10-yr risk **44% vs 5.8%** *Hall Blood 2003*
- **Acute event: therapeutic LMWH/UFH + start complement inhibition now** — anticoagulation alone does not stop it. Eculizumab cut events **7.37 → 1.07/100 pt-yr** *Hillmen Blood 2007*. Continue anticoagulation long term after a thrombosis
- **Primary prophylaxis: not routine on a complement inhibitor** — 21 patients without prior thrombosis stopped warfarin with no events *Kelly 2011*. Warfarin for clone >50% *Hall 2003* is the no-access fallback

AA / PNH OVERLAP

- **Small PNH clones are common in AA** — screen by high-sensitivity flow; a clone supports immune-mediated AA
- **Clones can expand after IST** into hemolytic PNH (or vanish) — re-check flow at least yearly
- **Severe marrow failure is treated as AA** (IST or transplant); complement inhibition only for clinically significant hemolysis
- **Clone size — rough guide, no cutoff defines PNH** (category = hemolysis + marrow status): **<10%** usually subclinical (typical in AA) · **10–50%** grey zone, check LDH/retic/haptoglobin · **>50%** typical of classic PNH, highest thrombosis risk

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COMPLEMENT CASCADE



B iptacopan: factor B inhibitor
D danicopan: factor D inhibitor (add-on)
P pegcetacoplan: C3 inhibitor
C5i eculizumab · ravulizumab · crovalimab

CD55 dismantles the C3 convertase; CD59 blocks the MAC. Both missing on PNH cells. C5i stop IVH, not C3b coating, so EVH persists. Blocking B, D or C3 stops both.

INFECTION SAFETY

- **MenACWY + MenB ≥2 wk before the first dose**; if treatment cannot wait, give antibiotic prophylaxis until 2 wk after vaccination. **Revaccinate per ACIP** for as long as the patient is on therapy
- **Proximal inhibitors (pegcetacoplan, iptacopan, danicopan) also block opsonization** → boxed warning extends to **S. pneumoniae and Hib**; vaccinate against pneumococcus (Hib per ACIP)
- **Vaccines do not fully protect** — meningococcal disease has occurred in vaccinated patients. Every patient carries the **safety card**
- **Fever or meningism = emergency:** blood cultures + empiric ceftriaxone immediately, then evaluate

PREGNANCY

- **High-risk for thrombosis, peaking postpartum**. Use **eculizumab or ravulizumab**; proximal agents lack human data — switch before conception
- Eculizumab, 75 pregnancies: **no maternal deaths**, 3 fetal deaths (4%); **dose/frequency increased in 54%**; LMWH in 88%; both thromboses postpartum *Kelly NEJM 2015*
- **Prophylactic LMWH** through pregnancy and the postpartum period; expect to escalate the C5i in the 2nd–3rd trimester

AGENTS — DOSING

Agent	Target · dosing	Notes
Eculizumab *Soliris 2007 + biosimilars*	**C5**, IV. **600 mg weekly ×4 → 900 mg wk 5 → 900 mg q2wk**	q2wk infusions; trough BTH. **C5 R885H** variant: no response
Ravulizumab *Ultomiris 2018*	**C5**, IV, weight-based load → maintenance 2 wk later, then **q8wk**: 40–<60 kg **2400 → 3000 mg** · 60–<100 kg **2700 → 3300** · ≥100 kg **3000 → 3600**. Switching: load 2 wk after last eculizumab	Default C5i; same R885H gap
Crovalimab *PiaSky 2024, ≥13 y + ≥40 kg*	**C5** (different epitope). **D1 1000 mg IV** (1500 if ≥100 kg) → **340 mg SC D2, 8, 15, 22 → 680 mg SC q4wk** from D29 (1020 if ≥100 kg). Switching: load at the next scheduled dose	**Switch from ecu/ravu → type III immune-complex reactions** (19% per label). Works in R885H
Pegcetacoplan *Empaveli 2021*	**C3**, SC. **1080 mg twice weekly**; every 3 d if LDH >2× ULN. From eculizumab: **overlap 4 wk**, then stop it; from ravulizumab: start ≤4 wk after last dose	Injection-site reactions. **Monitor ≥8 wk after stopping**
Iptacopan *Fabhalta 2023*	**Factor B**, oral. **200 mg BID**. Start ≤1 wk after last eculizumab or ≤6 wk after last ravulizumab	**Missed doses → hemolysis.** Monitor ≥2 wk after stopping
Danicopan *Voydeya 2024*	**Factor D**, oral, **added to** ravulizumab/eculizumab. **150 mg TID → 200 mg TID** if Hb rise <2 g/dL at 4 wk or transfused	**Never monotherapy** — for cs-EVH only

All six carry a boxed warning for serious meningococcal infection and are REMS-restricted. No approval yet for pozelimab + cemdisiran (investigational).

BREAKTHROUGH HEMOLYSIS (BTH)

- **BTH = return of IVH on therapy:** LDH up, dark urine, Hb falls, symptoms back
- **Pharmacokinetic** (C5i): drug runs out before the next dose — classic late in the eculizumab interval. Fix: shorten the interval / raise the dose, or move to ravulizumab *301: BTH 4.0% vs 10.7%*
- **Pharmacodynamic:** a complement-amplifying trigger — **infection, surgery, pregnancy**, vaccination. Treat the trigger; give an extra C5i dose
- **On a proximal inhibitor BTH can be abrupt and severe** — the whole, now-unprotected PNH RBC mass lyses at once. Check adherence, **give a C5i rescue dose** (eculizumab/ravulizumab), transfuse, involve a PNH center
- **Never stop a proximal inhibitor abruptly** — rebound hemolysis; bridge directly to another agent

KEY TRIALS

Trial	Result → practice
TRIUMPH *NEJM 2006*	Eculizumab vs placebo, transfused pts (n=87): Hb stabilized **49% vs 0%**; median 0 vs 10 units
SHEPHERD *Blood 2008*	Single-arm, broader population (n=97): LDH ↓87%; **51% transfusion-free** for 52 wk
Study 301 / 302 *Blood 2019*	Ravulizumab q8wk **noninferior** to eculizumab q2wk in naive (n=246) and switched (n=195) patients
COMMODORE 2 *AJH 2024*	Crovalimab vs eculizumab, naive (n=204): hemolysis control **79.3% vs 79.0%**, noninferior. COMMODORE 3 (China, single-arm) supportive
PEGASUS *NEJM 2021*	Switch from eculizumab, Hb <10.5 (n=80): pegcetacoplan **Hb +3.84 g/dL vs eculizumab**; transfusion-free 85% vs 15%
PRINCE *Blood Adv 2023*	Naive, vs supportive care (n=53): Hb stabilization **85.7% vs 0%**
APPLY-PNH *NEJM 2024*	Anti-C5 with Hb <10 → switch to iptacopan (n=97): Hb ↑≥2 g/dL in **51/60 vs 0/35**; Hb ≥12 in 42/60
APPOINT-PNH *NEJM 2024*	Naive, single-arm (n=40): Hb ↑≥2 g/dL in **31/33**; no transfusions
ALPHA *Lancet Haem 2023*	Danicopan added to C5i for cs-EVH: Hb **+2.94 vs +0.50 g/dL** at 12 wk (diff 2.44)

PEARLS

- **On a C5i the RBC clone grows** — PNH RBCs now survive. That is treatment effect, not progression; follow the granulocyte clone
- **DAT C3d+ on a C5i = EVH**, not a new autoimmune hemolysis
- **LDH on a C5i usually sits ≤1.5× ULN**; a sudden rise is BTH until proven otherwise
- **C5 R885H polymorphism** (described in Japanese patients): no response to eculizumab or ravulizumab — crovalimab binds a different C5 epitope
- Eculizumab/ravulizumab protect against thrombosis, but **after a thrombosis anticoagulation continues**